

JASC DATA PROCESSOR

Comprehensive User Manual

A lightweight, modular desktop application for data ingestion, visualization, mathematical transforms, statistics, and curve fitting.

Reference Manual & Guide

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1. Installation and First Launch

System requirements

- Python 3.10 or later.
- A desktop environment capable of displaying a Qt application.
- The packages in requirements.txt: PySide6, pandas, NumPy, Matplotlib, SciPy, odrpack, SciencePlots, and statsmodels.

Install dependencies

Open a terminal in the repository root and run:

```
python -m pip install -r requirements.txt
```

Using a virtual environment is recommended so that JASC's package versions remain isolated from other Python projects.

Start JASC

Run the application from the repository root:

```
python main.py
```

main.py is the supported entry point. Do not run ui/main_window.py directly. In the supplied VS Code workspace, the Run JASC Data Processor launch configuration starts the correct entry point.

At first launch, JASC opens an empty workspace. Create a sheet, import data, or open a .jasc project to begin.

File types at a glance

Purpose	Supported format
Tabular import	.csv, .txt
Tabular export	.csv, .txt / tab-delimited text
Project save/open	.jasc
Figure digitizer	PNG, JPEG, BMP, TIFF, GIF, and other image formats readable by Qt
Plot image export	Formats offered by the embedded Matplotlib toolbar

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2. Application Window, Menus, and Shortcuts

Main layout

```
+-------------------------------------------------------------------------+
| File | Edit | View | Data | Analysis | Plot                            |
+-------------------------------------------------------------------------+
| Main toolbar: project controls, new sheet, import, digitizer, export    |
+-------------------+---------------------------------+-------------------+
| Workspace         | Central work area               | Plot Inspector    |
|                   |                                 |                   |
| Datasets          | Data tab: worksheet editor      | Series            |
| Saved Plots       | Plot tab: Matplotlib canvas     | Axes              |
|                   | and navigation toolbar          | Legend            |
|                   |                                 | Locator           |
+-------------------+---------------------------------+-------------------+
| Status bar: active operation, warnings, progress, and result messages   |
+-------------------------------------------------------------------------+
```

- Workspace Navigator: The left dock contains the dataset and saved-plot lists. Select a dataset to display it in the Data tab. Select a saved plot to load it, subject to the unsaved-plot prompt.
- Data tab: Shows the selected worksheet and a summary containing its dimensions, memory estimate, metadata/formula counts, and backend.
- Plot tab: Shows the current plot canvas, the standard Matplotlib navigation toolbar, and plot-management buttons.
- Plot Inspector: The right dock is synchronized with the Plot tab and contains Series, Axes, Legend, and Locator controls. It is hidden while the Data tab is active.
- Status bar: Reports completed operations, selected outputs, large-data warnings, and batch progress summaries.

JASC automatically follows the operating-system light or dark palette and reapplies theme-aware table, input, dock, and dialog styling when the Qt palette changes.

View controls

The View menu can independently show or hide:

- Workspace Navigator: the complete left dock.
- Workspace Datasheets: the dataset section within the navigator.
- Saved Plots: the saved-plot section within the navigator.
- Plot Inspector: the right plot-control dock.

Use View > Reset View to Default to restore the normal dock arrangement and visibility.

Menu reference

Menu	Commands
File	Open Project, Save Project, Save Project As, New Blank Sheet, Close Project, Import Data, Import From Figure, Export Dataset, Exit
Edit	Undo, Redo, Delete Selected Rows, Delete Selected Columns
View	Navigator/section/inspector visibility and Reset View to Default
Data	Rename/Remove Dataset; Add/Insert Row or Column; Fill Range; all Data Preparation tools; Column Metadata; Apply Formula
Analysis	Derivative, Integral, Smoothing, Interpolation, FFT, Inverse FFT, Statistics, Custom Fit Models, Curve Fit
Plot	New Empty Plot, Save Current Plot, Load Selected Plot, Remove Selected Plot

Keyboard shortcuts

Shortcut	Action
Ctrl+O	Open project
Ctrl+S	Save project
Ctrl+Shift+S	Save project as
Ctrl+W	Close project
Ctrl+N	New blank sheet
Ctrl+I	Import data
Ctrl+E	Export selected dataset
Ctrl+Z	Undo worksheet edit
Ctrl+Y	Redo worksheet edit
Delete	Delete selected rows
Ctrl+C	Copy selected worksheet values
Ctrl+V	Paste worksheet values at the active cell
Ctrl+Shift+F	Apply Formula
Ctrl+Shift+C	Curve Fit
Ctrl+Shift+R	Reset view layout
Ctrl+Alt+N	New empty plot
Ctrl+Alt+S	Save current plot

---

3. Projects, Datasets, and the Workspace

Workspace concepts

- A dataset is a named pandas worksheet with a table, formulas, column metadata, source path, and backend label.
- A saved plot is a named snapshot of plot series, axes, legend, locator, style, and figure settings.
- A custom fit model is an expression-based fitting function stored with the project.
- The workspace is the collection of all three.

Names are made unique automatically. If Sheet already exists, another dataset requested with that name becomes Sheet (2), then Sheet (3), and so on. Saved plots and custom models use equivalent unique-name behavior.

Open, save, and close a project

1. Choose File > Open Project... to load a .jasc file. Opening a project replaces the current workspace after confirmation.
2. Choose File > Save Project to save to the current project path. The first save opens the Save As dialog.
3. Choose File > Save Project As... to choose a different path. JASC adds .jasc if no suffix is supplied.
4. Choose File > Close Project to clear the workspace without closing JASC. This action warns that unsaved changes will be lost.

Save before using File > Exit or closing the application window. Project-level automatic save and recovery are not implemented.

What a `.jasc` project preserves

A project is a ZIP archive containing CSV worksheet payloads and JSON metadata. It preserves:

- Dataset names, values, source paths, and backend labels.
- Column formulas.
- Column roles, value types, precision, units, long names, and comments.
- Operation-provenance comments written by JASC.
- Saved plot definitions, including plot series and appearance settings.
- Project-scoped custom fit models.

Because worksheet payloads are CSV, specialized pandas dtypes are reconstructed on load rather than serialized as native binary dataframe types.

Dataset navigation and management

- Select a dataset in the navigator to display it.
- Right-click a dataset and choose Rename... or Delete.
- The same actions are available as Data > Rename Dataset... and Data > Remove Dataset....
- Renaming a dataset updates references inside saved plots. The active plot draft is cleared so it cannot retain stale controls.
- Deleting a dataset asks for confirmation. Dataset deletion is not part of worksheet Undo/Redo.

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4. Importing and Creating Data

Import CSV or TXT data

1. Choose File > Import Data... (Ctrl+I) or use the toolbar action.
2. Select a .csv or .txt file.
3. In Import Options, choose one delimiter:
- Comma
- Semicolon
- Tab
- Space / whitespace (one or more whitespace characters)
4. Check First row contains column names when row 1 contains headers. Clear it to generate Column 1, Column 2, and so on.
5. Confirm the import.

The current import dialog does not auto-detect a delimiter or header row, does not select an arbitrary starting row, and always reads text as UTF-8. If decoding fails, convert the source file to UTF-8 before importing.

The imported dataset uses the source filename as its base name and remembers the source path. Files of at least 100 MiB trigger a warning before parsing. Large parsed worksheets trigger a second size/memory warning before they enter the workspace.

Create a blank worksheet

Choose File > New Blank Sheet (Ctrl+N), the toolbar action, or New Sheet in the dataset navigator. JASC creates a normal workspace dataset with:

- 25 initially blank rows.
- Numeric x and y columns.
- X and Y column roles already assigned.

The sheet is immediately editable in the Data tab. Use Data menu commands to add or insert rows and columns.

Fill an arithmetic range

Choose Data > Fill Range... to populate a sequence. Configure:

- Target column: an existing or new column.
- Start row: one-based worksheet row.
- Count: number of values.

- Start value and Step.

JASC grows the worksheet if required and writes start + offset * step. It marks the output Numeric, assigns the X role when the name is x, time, or t, removes a stale formula from the overwritten column, and records provenance in the column comment.

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5. Digitizing Data from Figure Images

Use File > Import From Figure... to extract calibrated data from PNG, JPEG, BMP, TIFF, or GIF figures. The digitizer supports optional perspective correction, linear/log calibration, manual points and uncertainties, automatic curve tracing, ROI restriction, connected-symbol detection, and sampled-template symbol matching.

Canvas controls and tool state

The toolbar above/below the image provides Load Image, Fit View, 1:1, Zoom In, Zoom Out, Pan, and Magnifier. The mouse wheel zooms around the pointer. The Precision Zoom panel follows the cursor and shows an enlarged neighborhood with pixel and calibrated-coordinate readouts.

The yellow mode banner reports what the next canvas click or drag will do. It is a status display, not a button. Use the controls in the Correction, Calibration, and Extraction tabs to activate a picking tool. Enable Pan when dragging should move the image without picking data.

Step 1: optional perspective correction

Use the Correction tab when a photograph or scan is skewed.

1. Click Pick top-left, Pick top-right, Pick bottom-right, and Pick bottom-left, clicking the corresponding plot-frame or axis-bar corners in that order.
2. Review the green quadrilateral.
3. Set the corrected image width and height in pixels.
4. Click Apply Correction.

Calibration and extraction then operate on the rectified working image. Clear Points removes the four correction points. Reset Original restores the original image and clears pixel-dependent calibration and extraction state.

Step 2: calibrate X and Y

The calibration maps pixels to data coordinates using two references per axis.

1. In Calibration, select X axis point 1, click its pixel location, and enter the known X value.
2. Repeat for X point 2, Y point 1, and Y point 2.
3. Choose Linear or Log10 independently for X and Y.
4. Review the calibration status. Use Clear Calibration to start over.

Calibration values must be distinct on each axis. Log10 calibration requires positive reference values and is appropriate only for logarithmic axes.

Step 3A: manual point extraction

1. Open Extraction and click Manual Mode.
2. Click data points on the image. Use Stop Picking when finished.
3. Use Undo Last to remove the latest manual point or Clear Manual to remove all manual points.

To attach uncertainty to a manual point:

1. Select the target point number in Error target point.
2. Click Radius Error, then click one radial boundary. This assigns symmetric pixel uncertainty from the distance to the point.
3. Alternatively, click 4-Point Error and pick the X-low, X-high, Y-low, and Y-high boundaries.

Manual uncertainties appear as blue error bars and are exported in both pixel and calibrated data units.

Step 3B: automatic curve or line tracing

The Automatic Extraction Region is shared by curve tracing and symbol detection.

1. Optionally click ROI Brush and paint the region to scan. Adjust Brush size as needed. Without ROI marks, JASC scans the calibrated plot area.
2. In the Curves/Lines subtab, choose Dark pixels or Selected color.
3. For dark pixels, adjust Dark threshold.
4. For color tracing, use Pick Color... or Sample From Image, then set Color tolerance.
5. Set X pixel step to control horizontal sampling density.
6. Click Run Auto Trace. Use Clear Auto to discard the trace.

Step 3C: symbol detection

The Symbols subtab has two related paths:

- Detect Symbols finds connected foreground components using the current dark/color rule and ROI. Use Min area, Max area, and Max symbols to reject lines, noise, or oversized objects.
- Detect Sampled matches a learned marker template. Click Sample Symbol, then click near a marker center for radius-based capture or drag a box around it. Configure the shape prior, sample radius, minimum match score, minimum separation, and maximum symbol count before detection.

Shape priors are Auto from sample, Dot / circle, X cross, Plus cross, Square, Diamond, Triangle up, and Triangle down. They help suppress line pixels touching a marker. Clear Template discards the learned template; Clear Symbols discards detected centers.

Symbol outputs include area, width, height, match score when applicable, template dimensions, and uncertainty estimated from marker size.

Step 4: review and create the dataset

Open Output to set the dataset name and inspect the read-only preview. Each row can include:

- Calibrated X, Y, dX, and dY.
- Pixel X, Pixel Y, dPx, and dPy.
- Extraction method, trace rule/color, ROI count, and source image.
- Symbol ID, area, score, size, and template diagnostics.

Click Create Dataset. JASC requires a loaded image, complete four-point calibration, and at least one manual, traced, or symbol point. The created dataset receives numeric coordinate metadata, X/Y/error roles where appropriate, source/provenance comments, and the image source path.

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6. Editing Worksheets and Column Metadata

Select and edit cells

- Double-click a cell, enter a value, and press Enter to commit.
- Blank input is stored as a missing value.
- Numeric/Scientific metadata rejects non-numeric text. Text columns preserve entered text.
- Select complete rows or columns using their headers; use Ctrl/Shift selection where supported by the platform.

Add, insert, and delete structure

- Data > Add Row appends a blank row.
- Data > Insert Row inserts before the active/selected row.
- Data > Add Column... appends a uniquely named blank column.
- Data > Insert Column... inserts before the active/selected column.
- Edit > Delete Selected Rows or Delete removes every selected row.
- Edit > Delete Selected Columns removes every selected column.

Row and column edits participate in Undo/Redo when the worksheet fits within the history budget.

Clipboard operations

1. Select cells and press Ctrl+C, or right-click and choose Copy Values.
2. Activate the destination cell and press Ctrl+V, or choose Paste Values.

Values use tab-delimited clipboard text. JASC expands the worksheet to fit the pasted rectangle, infers numeric blocks where safe, preserves explicit Text destinations, and clears formulas on overwritten columns.

Right-click selected columns and choose Copy Column(s). In another JASC worksheet, right-click a destination column and choose Paste Column(s). Whole-column copy uses a JASC-specific clipboard payload that preserves:

- Column headers.
- Values.
- Column roles, types, precision, units, long names, and comments.

Pasted columns are inserted before the destination and renamed automatically if their names collide. The metadata-preserving payload is available only while the clipboard still contains the JASC column data; plain-text applications receive the tabular values.

Undo and redo

Use Edit > Undo (Ctrl+Z) and Edit > Redo (Ctrl+Y). History covers cell edits, row/column changes, clipboard pastes, Fill Range, formula writes, metadata/column renames, and current-sheet calculus/smoothing changes where the operation captures editor state.

History is dataset-specific and resets when the displayed dataset changes. Each undo and redo stack retains at most 20 snapshots and approximately 256 MiB. If one worksheet snapshot exceeds the budget, JASC performs the edit but warns that it cannot be undone. Creating/removing datasets or batch changes to non-visible sheets are not guaranteed to be undoable from the current table history.

Column Metadata dialog

Open Data > Column Metadata..., double-click a column header, or use the table context menu. Each row exposes:

Field	Purpose
Column name	Renames the column; names must be nonblank and unique
Role	Hints default selections in plotting, fitting, interpolation, and analysis
Type	Controls display and explicit numeric/text validation
Precision	Auto or 0-15 displayed decimal places; Scientific uses exponential notation
Unit	Included in suggested plot-axis labels
Long name	Human-readable axis/header description
Comment	Notes, source details, and generated-operation provenance

Available roles are None, X, Y, Z, Error, X error, Y error, Group, Label, and Weight. Available types are Auto, Numeric, Scientific, and Text.

Changing display precision does not change stored values. Renaming a column carries its formula and metadata forward and updates active and saved plot column references.

Formula and provenance indicators

Formula-backed and metadata-rich column headers expose details in tooltips. Generated operations append an Operation: line to column comments, replacing the prior generated-operation line while retaining other user comments.

---

7. Data Preparation

Data Preparation commands create new result datasets. The source sheet remains unchanged. Matching metadata is copied where output columns retain their original names, and JASC records the operation in output-column comments.

If an operation returns no rows, JASC asks whether to create the empty output sheet.

Filter / Query

Choose one dataset, one column, one condition, and an output sheet. Supported conditions are:

- Equals / does not equal.
- Greater than, greater than or equal, less than, less than or equal.
- Between two numeric values, inclusive.
- Contains / starts with.
- Is in a comma-separated list.
- Is blank / is not blank.

Numeric comparison text must parse as a number. The current dialog applies one condition at a time; multi-condition query building and in-place filtering are not implemented.

Sort

Select one or more columns, Ascending or Descending, and an output name. Multiple keys are applied in their worksheet order using a stable sort.

Remove Duplicates

Select the columns that define a duplicate group and choose:

- Keep first occurrence.
- Keep last occurrence.
- Remove every row in a duplicate group.

Blank/NaN values in the same selected key columns are treated as matching.

Missing Values

Select one or more columns and a strategy:

- Drop rows missing a selected value.
- Fill with a literal value.
- Fill numeric blanks with the column mean.
- Fill numeric blanks with the column median.
- Interpolate numeric gaps in both directions.

Mean, median, and interpolation coerce the selected output columns to numeric values; non-numeric entries become missing before filling.

Bin / Average

Choose a numeric bin column, one or more value columns, and a bin count. JASC creates equal-width bins between the observed minimum and maximum and reports:

- Bin interval, start, end, and center.
- Row count.
- Mean, sample standard deviation, SEM, minimum, and maximum for every selected value column.

Binning requires a non-constant numeric bin column.

Group / Aggregate

Choose one or more group columns, distinct numeric value columns, and one or more aggregations: count, mean, standard deviation, SEM, minimum, maximum, and sum. Missing group keys are retained as groups.

Merge / Join

Choose left and right datasets, one key from each, and Inner, Left, Right, or Outer join. Same-named right-side columns receive a _right suffix where required.

Reshape

- Transpose converts columns to rows and adds a source_column field.
- Melt / stack retains selected identifier columns and stacks selected value columns into variable and value fields.

---

8. Formulas and Expressions

Choose Data > Apply Formula... (Ctrl+Shift+F) to calculate a new column or update an existing column with a constrained Python-like expression.

Basic workflow

1. Select or type a Target column.
2. Choose what happens if the target exists:
   - Create a unique new column.
   - Stop and warn.
   - Overwrite/update the existing column.
3. Enter the expression.
4. Optionally apply only to selected rows. Selected-row mode applies only to the current sheet.
5. Select one or more sheets for a full-column/batch operation.
6. Choose exact-name or position column matching for batch targets.
7. For formula batches, choose Auto, Sequential, or Parallel processes, plus the maximum worker count.
8. Confirm the preflight preview and run.

When a formula-backed target column is selected, JASC reloads its stored expression unless doing so would overwrite an expression currently being edited.

Referencing columns and rows

Syntax	Meaning
voltage	A column whose name is a safe Python identifier and does not conflict with a built-in name
col1, col2, ...	First, second, and later columns by position
col("Voltage (V)")	Exact-name access for spaces, punctuation, keywords, or conflicting names
row	One-based row-number series
index	The pandas index as a series

Use the dialog's Insert colN and Insert col("Name") buttons to avoid typing mistakes.

Examples:

```
col("Voltage (V)") / col("Current (A)")
np.sin(col1) * np.exp(-col2)
where(col1 > 0, sqrt(col1), nan)
row * 0.1
special.erf((col1 - 5) / 2)
signal.savgol_filter(col2, 11, 3)
np.random.normal(0 * col1, 1)
```

Available safe namespaces

The evaluator exposes a curated subset, not unrestricted NumPy/SciPy or Python.

- NumPy: functions are available both directly (sin, sqrt, where) and through np (np.sin, np.sqrt, np.where). Categories include arithmetic, trigonometry, logarithms, rounding, reductions, cumulative operations, gradients/differences, clipping, conditions, interpolation, finite-value checks, and array helpers.
- NumPy subnamespaces: selected np.fft functions, np.linalg.norm, and selected np.random generators.
- SciPy aliases: scipy and sp.
- SciPy subnamespaces: special, signal, stats, and integrate, also accessible as scipy.special, scipy.signal, scipy.stats, and scipy.integrate.
- Constants: pi, e, nan, and inf, also available through np.

The Help panel inside Apply Formula lists the allowed functions and examples for the installed version.

Safety and expression rules

- Use ** for powers. JASC rejects ^ because Python interprets it as bitwise XOR.
- File access, imports, Python builtins, private attributes, assignments, comprehensions, and arbitrary object access are blocked.
- A scalar result is repeated across the sheet. A vector result must match the worksheet length.
- Numeric/Scientific metadata is validated before downstream math operations.
- Applying a formula stores the expression in project metadata and records it in the target column comment.

---

9. Calculus, Smoothing, FFT, and Interpolation

All workbenches in this section support multiple compatible worksheets. Derivative, Integral, and Smoothing write columns into target worksheets; FFT, Inverse FFT, and Interpolation create result datasets.

Derivative

Choose Analysis > Derivative..., select one X column and one or more Y columns, then choose:

- NumPy gradient.
- Forward difference.
- Backward difference.
- Central difference.
- Savitzky-Golay derivative.

Set a target column or template, conflict policy, matching mode, target sheets, and whether to plot immediately. Use {column} in the target name when processing several Y columns.

X/Y data are aligned after numeric conversion and blank removal. X values cannot contain duplicate positions. The Savitzky-Golay derivative also requires evenly spaced X values.

Integral

Choose Analysis > Integral.... Methods are:

- Cumulative trapezoid.
- Cumulative Simpson.
- Left rectangle.
- Right rectangle.
- Midpoint rectangle.

The Initial value is the integration constant added to the cumulative result. Integral has the same multiple-Y, target, conflict, batch, and immediate-plot controls as Derivative.

Smoothing

Choose Analysis > Smoothing.... The dialog previews the first selected Y signal and limits preview rendering to at most 4,000 evenly sampled points.

Method	Relevant controls
Adjacent average	Window points

Moving median	Odd window points
Savitzky-Golay	Odd window points and polynomial order smaller than the window
LOWESS / LOESS	Fraction of data and robust iterations
FFT low-pass	Cutoff as percent of Nyquist; evenly spaced X required
Derivative smoothing + integration	Derivative smoother, optional odd slope-spike median prefilter, endpoint preservation

Derivative reconstruction calculates dy/dx, optionally removes derivative spikes, applies any non-recursive smoothing method, then integrates from the original first Y value. Preserve first and last data values distributes residual drift so the reconstruction ends at the original last value.

The default target template is {column}_smoothed. Choose a conflict policy, name/position matching, target sheets, and whether to plot original versus smoothed data immediately.

FFT

Choose Analysis > FFT... and select one or more signal columns. Choose an X/time column to derive sample spacing, or Sample index for spacing 1. Available windows are:

- None / rectangular.
- Hann.
- Hamming.
- Blackman.
- Bartlett.
- Flat top.

When an X/time column is used, values must be finite and evenly spaced within the operation tolerance. Each result dataset contains frequency, real, imaginary, magnitude, and phase. The optional immediate plot shows magnitude versus frequency.

Inverse FFT

Choose Analysis > Inverse FFT... and select one or more real spectrum columns. Imaginary input may be:

- Empty, meaning zero imaginary component.
- One column reused for every selected real column.
- The same number of columns as real selections, paired by selection order.

Each result contains sample, real, imaginary, and magnitude. The optional immediate plot shows the reconstructed real signal.

Interpolation

Choose Analysis > Interpolation.... Select one or more value columns and a dimensional mode.

Mode	Methods
1D	Linear, nearest neighbor, cubic spline, PCHIP, Akima, smoothing spline
2D	Nearest neighbor, linear scattered, cubic spline, piecewise cubic / Clough-Tocher, smooth bivariate spline, RBF linear, thin plate, cubic, Gaussian
N-D	Nearest neighbor, linear scattered, RBF linear, thin plate, cubic, Gaussian

N-D mode requires at least two unique coordinate columns. Value columns must differ from coordinate columns. Duplicate source coordinates are averaged.

- Generated range/grid: available for 1D and 2D. Set start, end, and count for each dimension. A 2D query produces the Cartesian X/Y grid.
- From existing columns: choose one query column per coordinate dimension.
- Manual coordinates: enter one point per line, separating values with spaces, commas, or tabs.

Generated queries are limited to 1,000,000 points.

- Outside range applies to 1D: blank outside the source range, extrapolate where supported, or clamp to edge values.
- Smoothing is enabled for smoothing splines and RBF methods.
- RBF epsilon is enabled for Gaussian RBF; zero displays as automatic.

The result preserves query coordinates and adds one {value_column}_interpolated prediction column per selected value. Result-sheet templates can use {source} and {column} in batch mode.

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10. Statistics Workbench

Choose Analysis > Statistics.... Select an operation, its primary/secondary/group inputs, output name, target sheets, column-binding mode, and optional companion plot. Numeric operations ignore blanks, non-numeric values, and infinities after validating explicit metadata.

Single-column operations

- Complete report.
- Descriptive summary: count, missing count, mean, median, standard deviation, variance, SEM, min, max, range, and coefficient of variation.
- Quantiles and percentiles.
- Distribution shape: skewness and kurtosis.
- Normality tests: Shapiro-Wilk, D'Agostino-Pearson, Jarque-Bera, and Anderson-Darling.
- Outlier diagnostics using Z-score and IQR fences.
- Mean confidence interval.
- Z-score result series.
- Cumulative statistics result series.

Two-column operations

Two-column operations can reuse one secondary column for several primary columns or pair equal-sized selections by order. Paired analyses align rows before dropping incomplete pairs.

- Complete comparison report and pair summary.
- Pearson, Spearman, and Kendall correlations.
- Covariance matrix values.
- Linear regression.
- Error metrics: MAE, RMSE, mean error, and R-squared against the primary column.
- Bland-Altman agreement.
- Paired t-test and Wilcoxon signed-rank.
- Welch independent t-test, Mann-Whitney U, and two-sample Kolmogorov-Smirnov.
- Levene and Bartlett variance tests.

Multiple-column operations

Select two or more columns from the primary sheet:

- Complete multi-column report.
- Descriptive summary.
- Pairwise correlation matrix tests.

- Covariance matrix values.
- Principal component analysis with eigenvalues, explained variance, loadings, and score rows.

Grouped operations

Choose one or more primary numeric columns and exactly one group/factor column, which may come from another sheet:

- Complete grouped report.
- Descriptive summary by group.
- Assumption checks.
- One-way ANOVA.
- Kruskal-Wallis.
- Tukey HSD post-hoc comparisons.
- Pairwise Mann-Whitney comparisons with Holm-Bonferroni adjustment.

Companion plots

Available choices depend on operation scope:

Scope	Plot choices
Single	Auto, histogram + normal curve, box, violin; Z-score/cumulative operations can also plot the result series
Pair	Auto, scatter, scatter + regression line, comparison box, comparison violin, Bland-Altman
Multi	Auto, selected-column box/violin; correlation heatmap for correlation/complete reports; PCA explained variance for PCA/complete reports
Group	Auto, box by group, violin by group

Batch statistics creates long summary results and, when applicable, a wide summary dataset. At most one companion plot is queued for a batch run so a large batch does not replace the plot repeatedly.

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11. Curve Fitting and Custom Models

Standard curve fitting

Choose Analysis > Curve Fit... (Ctrl+Shift+C).

1. Select one X column and one or more Y columns.
2. Select a model and optimizer.
3. Edit each parameter's initial value, optional lower/upper bounds, and Fixed checkbox.
4. Configure uncertainty, confidence, residual, and output options.
5. Select target sheets and column matching.
6. Confirm the batch preview and run.

Built-in models

- Linear, quadratic, and cubic.
- Exponential growth, exponential decay, and double exponential decay.
- Gaussian, asymmetric Gaussian, Lorentzian, and pseudo-Voigt.
- Sine and damped sine.
- Logistic and Gompertz.
- Error-function step.
- Power law.
- Log-normal peak.
- Weibull saturation.
- Heidler impulse.
- Bessel J and Bessel Y.
- Stretched exponential.

The dialog shows the selected equation, parameter order, and static defaults.

Optimizers and parameter constraints

- Levenberg-Marquardt is suitable for unconstrained least squares.
- Trust Region Reflective is the default and supports bounds.
- Dogbox supports bounded least squares.
- Orthogonal Distance Regression supports uncertainty in both X and Y.

Blank bounds are unbounded. Fixed parameters remain at their initial values. At least one parameter must remain free.

Uncertainty and weighting

- Select an optional X weight/sigma column.
- Select zero, one, or one-per-Y Y weight/sigma columns.
- Choose Sigma / standard deviation or Weight (larger = stronger).
- Treat sigma as absolute controls covariance scaling.

Selecting an X uncertainty column automatically switches to ODR and locks the optimizer control. Error/sigma values must be positive and finite where used.

Outputs and report

Configure:

- Confidence band on/off and confidence percentage, 50.0-99.9%.
- Number of generated fit-curve points, 10-10,000.
- Result sheet name.
- Residual sheet on/off and name.
- Residual plot on the right Y axis.
- Optional long-format parameter covariance/correlation sheet.

The Fit Report includes model/method, parameter estimates, fixed/free state, bounds, standard errors, covariance, R-squared, adjusted R-squared, RMSE, MAE, residual mean/standard deviation, point counts, and warnings. Successful immediate plotting queues raw data, fitted curve, optional confidence band, and optional residuals.

Custom fit models

Choose Analysis > Custom Fit Models... to Add, Edit, Duplicate, or Remove project-scoped models.

A custom model defines:

- A unique name.
- An expression using x as the independent variable.
- Comma-separated parameter names.
- One default initial value per parameter.

Example:

```
amplitude * np.exp(-x / tau) + offset
```

Parameter names must be unique Python identifiers, cannot be x, and cannot conflict with built-in model names. The validator exposes a curated selection of NumPy plus scipy.special, scipy.signal, scipy.stats, and scipy.integrate; imports, file access, arbitrary attributes, and ^ powers are blocked. Click Validate before saving when you want an explicit check.

Custom models appear in Curve Fit as Custom: model name and are stored inside the .jasc project.

---

12. Plotting and the Plot Inspector

The Plot tab holds one editable plot draft. A plot can contain multiple queued series, including series whose X, Y, and Z values come from different datasets.

Build and edit a plot

1. Open the Plot tab.
2. In Series, choose a plot family and subtype.
3. Configure the data mappings and type-specific appearance.
4. Click Add as New.
5. Repeat to build a multi-series plot.

Selecting a queued series loads it into the editor. Changes to its subtype, mapping, or appearance update the queued state and participate in Auto preview. Apply to Selected performs an explicit update. Use Remove Selected or Clear All to edit the queue.

Plot types and mappings

Family	Subtype	Data mapping and special controls
XY / Categorical	Line	X and Y; line style/width; optional left/right Y axis
XY / Categorical	Scatter	X and Y; marker and size; optional left/right Y axis
XY / Categorical	Line + Markers	X and Y; line and marker controls; optional left/right Y axis
XY / Categorical	Errorbar	X/Y plus optional X-error and Y-error columns
XY / Categorical	Vertical Bar	Numeric positions or text categories plus values; bar width and fill alpha
XY / Categorical	Horizontal Bar	Numeric positions or text categories plus values; bar width and fill alpha
XY / Categorical	Area	X/Y filled area, line controls, fill alpha
Distribution	Histogram	One value column, 1-1,000 bins, density, cumulative, fill alpha
Distribution	Box Plot	One or more value columns; mean, median, outlier controls
Distribution	Violin Plot	One or more value columns; mean, median, extrema, statistic-line width
Composition	Pie	Label column plus positive value column

Composition	Donut	Label column plus positive value column
Composition	Voronoi Composition	Label/value columns and Circular, Rectangular, or Polygonal container
2D Maps	Interpolated Heatmap	X/Y/Z scattered coordinates, interpolation method, grid resolution
2D Maps	Contour	X/Y/Z scattered coordinates, interpolation method, grid resolution
2D Maps	Filled Contour	X/Y/Z scattered coordinates, interpolation method, grid resolution

2D-map interpolation choices are Linear, Cubic spline, and Piecewise cubic. Maps require numeric Z values and at least three distinct non-collinear X/Y coordinate points. Rendering failures appear as a message on the canvas instead of a blank plot.

Composition plots cannot be mixed with XY series, and only one composition series can render at a time. Pie/donut require positive values. Voronoi regions are value-weighted approximations and automatically reduce label density for large category counts.

Common series appearance

Depending on plot type, the inspector exposes:

- Custom legend label and per-series Show this series in legend.
- Solid, dashed, dotted, dash-dot, or no line.
- Circle, square, triangle, diamond, cross, or dot markers.
- Automatic or named colors.
- Line width, marker size, fill alpha, and bar width.
- Left or right Y-axis assignment.

Column roles preselect likely mappings. Axis-label fields prefill from Long Name and Unit metadata while they remain blank or automatically managed; manually typed labels are not overwritten.

Axes, style, and figure

The Axes tab controls:

- Title; X, left-Y, and right-Y labels.
- Linear or log scale independently on X, left Y, and right Y.
- Optional minimum/maximum for each axis.
- Grid visibility.
- Physical figure width (3-20 inches) and height (2-15 inches).
- Style preset: Matplotlib default, Science, IEEE, Nature, or Notebook.
- Color cycle: Default, Bright, or High visibility.

- LaTeX text on/off.
- Custom comma-separated SciencePlots style tokens.

Log ranges must be positive. Invalid or reversed ranges mark the preview out of date without interrupting typing with modal dialogs. SciencePlots styles require the dependency in requirements.txt; LaTeX rendering also depends on a working external LaTeX installation.

Legend

The Legend tab controls:

- Visibility.
- Automatic or standard inside positions.
- Outside right, outside top, or outside bottom placement with reserved export space.
- Text size, optional title, and title size.
- Column count.
- Frame visibility and opacity.
- Marker scale and handle length.
- Whether locator markers and confidence bands appear.

Entries follow queued-series order across both Y axes, and duplicate labels are removed.

Auto preview and Apply

Auto preview is enabled by default and debounces global style/axis changes to avoid excessive redraws. Turn it off for expensive surfaces or large datasets, then click Apply when ready. The preview indicator reports current, pending, out-of-date, invalid, or failed state.

Data locators

Open Locator, select a queued series, and choose a lookup:

- Find Y from X interpolates Y on an ordinary XY series.
- Find X from Y finds every piecewise-linear crossing on an ordinary XY series.
- Surface plots use entered X and Y to interpolate Z.

Choose marker shape, size, and color, then click Add Locator. Located values appear in the list and on the correct left/right axis. Non-monotonic curves can return several X crossings; repeated shared vertices are deduplicated. A horizontal segment at the requested Y is reported as ambiguous instead of returning an arbitrary X.

Matplotlib canvas toolbar

Use the embedded Matplotlib toolbar for home/reset, back/forward, pan, zoom, subplot configuration where offered, and saving the rendered figure. This exports an image; it does not save editable JASC plot state.

Saved plots and unsaved drafts

- New / Empty Plot is the default unsaved workspace entry.
- Use the Plot menu or top-row buttons to create, save, load, and remove plot definitions.
- Saving under an existing name asks before replacement.
- Starting or loading another plot while the current draft has unsaved series changes offers Save, Discard, and Cancel.
- Analysis-generated plots detach from a previously loaded saved plot so they cannot silently overwrite it.

Saved definitions preserve queued series, locators, mappings, axes, legend, styles, figure dimensions, and type-specific settings. Save the overall .jasc project to persist them between application sessions.

---

13. Batch Processing

Batch-capable workbenches are:

- Apply Formula.
- Derivative and Integral.
- Smoothing.
- FFT and Inverse FFT.
- Interpolation.
- Statistics.
- Curve Fit.

Standard batch workflow

1. Configure the operation using the current sheet as the reference.
2. Select one or more entries in Apply to sheets.
3. Select Exact name or Column position matching.
4. Configure output names/conflict behavior where offered.
5. Confirm the operation dialog.
6. Review Batch Preflight Preview, including resolved inputs, outputs, status, and error details.
7. Click Run Batch.
8. Monitor the cancelable progress dialog.

Preflight rows marked failed or blocked explain incompatible columns, invalid mappings, or output conflicts before execution. Compatible jobs can still run when other rows fail, depending on the operation's plan.

Column binding

- Exact name resolves each requested header literally on every target sheet. Use it for standardized worksheets.
- Column position treats the reference mapping as col1, col2, and so forth and resolves the same positions on every target sheet. Use it for repeated exports whose headers differ but layouts match.

Column roles influence initial dialog choices but do not replace batch binding.

Output naming and collision policies

Where the dialog supports them:

- {source} expands to the source sheet name.
- {column} expands to the processed signal/value column.
- Create unique protects existing columns by generating a noncolliding name.

- Stop and warn blocks the colliding output.
- Overwrite/update permits replacement after the preflight makes the target explicit.

Dataset-producing operations also create unique names when a requested result name already exists.

Multiple input columns

- Derivative, Integral, Smoothing, FFT, Statistics, and Curve Fit can process multiple selected signal/value columns.
- Inverse FFT accepts zero, one shared, or one-per-real imaginary column.
- Curve Fit accepts zero, one shared, or one-per-Y sigma/weight column.
- Pair statistics accept one shared secondary column or one-per-primary secondary columns.
- Interpolation accepts multiple value columns.

Parallel formula execution

Apply Formula uniquely exposes Auto, Sequential, and Parallel processes plus maximum workers. Auto decides whether parallel processing is worthwhile for the headless target sheets. The visible current sheet may be handled safely in the GUI path while compatible background jobs use worker processes.

Progress, cancellation, and summaries

Batch operations display a modal progress dialog and check for cancellation between jobs/results. A cancellation does not undo jobs already completed.

Multi-job runs create operation-specific summary datasets such as:

- _formula_batch_summary
- _derivative_batch_summary / _integral_batch_summary
- _smoothing_batch_summary
- _fft_batch_summary / _ifft_batch_summary
- _interpolation_batch_summary
- _statistics_batch_summary and sometimes _statistics_batch_wide_summary
- _fit_batch_summary

Summaries include source sheets/columns, resolved mappings, requested outputs, status/errors, execution details, and operation-specific metrics.

---

14. Export, Persistence, and Large Data

Export a dataset

Select a dataset and choose File > Export Dataset... (Ctrl+E). Choose CSV or tab-delimited text. JASC exports column headers and values without the pandas row index.

Dataset export does not include formulas, column metadata, saved plots, or custom models. Save a .jasc project when those need to be preserved.

Export a plot

Use the save button in the Matplotlib toolbar. The available raster/vector formats depend on Matplotlib. Outside legends reserve figure space in the exported output.

An exported image is static. Save the plot definition and project when later editing is required.

Large-data thresholds and behavior

- Delimited files of at least 100 MiB trigger a pre-import warning.
- A worksheet is flagged as large at 2,000,000 cells or an estimated 250 MiB of DataFrame memory.
- Column sizing uses at most 100 sampled rows normally and 25 rows above the 250,000-cell responsiveness threshold.
- Clipboard parsing, worksheet growth, range fill, and rectangular assignment use block-oriented operations.
- .jasc worksheet CSV entries stream from the ZIP archive during project load.
- Undo/redo retains at most 20 snapshots and approximately 256 MiB per stack.

All datasets are currently held in memory by pandas. CSV/TXT import and much project parsing still run synchronously, so very large files can pause the interface even after confirmation. Background import, paged/out-of-core datasets, autosave, and recovery are not implemented.

---

15. Troubleshooting and Current Limitations

JASC does not start

Run from the repository root and verify the active interpreter:

```
python --version
python -m pip install -r requirements.txt
python main.py
```

If python is not found, activate the environment where the dependencies were installed or invoke that environment's Python executable directly.

A CSV/TXT file parses incorrectly

- Reimport and choose the correct delimiter.
- Check or clear First row contains column names as appropriate.
- Save the source as UTF-8.
- JASC does not currently auto-detect delimiters/headers, skip arbitrary leading rows, preview imports, or choose encodings.

A numeric operation rejects a column

- Open Column Metadata and confirm it is Auto, Numeric, or Scientific rather than Text.
- Remove or correct non-numeric text in explicitly Numeric/Scientific columns.
- Check for duplicate X values, missing aligned points, or uneven spacing where the selected method requires uniform samples.

A formula fails

- Use **, not ^, for powers.
- Use col("Exact Header") for spaces, punctuation, keywords, or names that conflict with functions.
- Check the Apply Formula Help panel for the curated function list.
- Ensure a vector expression returns exactly one value per worksheet row.

A fit does not converge

- Use realistic initial guesses.
- Confirm lower bounds are smaller than upper bounds and do not exclude the solution.
- Try Trust Region Reflective or Dogbox for bounded fits.

- Inspect missing, nonfinite, or nonpositive uncertainty values.
- Reduce the number of free parameters or fix known parameters.
- Confirm the model domain is valid for the selected X range, especially log-normal, power-law, Bessel Y, or fractional-power models.

A 2D map is blank or reports an error

- Choose distinct X, Y, and Z mappings.
- Provide at least three distinct, non-collinear coordinate points.
- Ensure Z is numeric and enough finite points remain after blank removal.
- Try Linear before cubic methods and reduce grid resolution for expensive data.

A plot style or LaTeX fails

- Reinstall requirements to provide SciencePlots.
- Disable Use LaTeX text if a LaTeX distribution is not installed.
- Remove unsupported tokens from Custom styles.

Current scope limitations

The current version does not provide:

- Excel, database, JSON/XML, HDF5, NetCDF, MATLAB, or instrument-format connectors.
- Import preview, arbitrary header/start rows, encoding selection, or batch file import.
- A workbook/folder hierarchy, matrix sheets, rich cell formatting, notes, or dependency/recalculation graphs.
- Multi-condition filters, pivot tables, resampling, date/time tooling, or reusable preparation recipes.
- Multi-panel pages, true 3D plots, polar/ternary plots, annotation objects, or general graph templates.
- Global multi-dataset fitting, automated model ranking, peak deconvolution, or surface fitting.
- A scripting console, macro recorder, plugins, autosave/recovery, installer, or true out-of-core data backend.

Consult Roadmap.md for planned work; roadmap entries are not instructions for currently available features.

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16. Quick Reference

Recommended end-to-end workflow

1. Import/create data.
2. Assign column names, roles, types, units, and long names.
3. Clean data with non-destructive Data Preparation outputs.
4. Apply formulas or analysis operations, using preflight for batch work.
5. Inspect generated comments and batch summaries.
6. Build an unsaved plot draft, configure axes/legend/locators, and save the plot definition.
7. Export final data/figures as needed.
8. Save the .jasc project to retain the editable workspace.

Output type summary

Operation	Primary output
Fill Range / Formula / Derivative / Integral / Smoothing	Column(s) written into target worksheet(s)
Data Preparation	New non-destructive dataset
FFT / Inverse FFT / Interpolation	New result dataset(s)
Statistics	New report/result dataset(s), optional plot
Curve Fit	Fit dataset, report, and optional residual/covariance/plot outputs
Figure digitizer	New calibrated dataset with diagnostics
Plot Inspector	Editable plot draft; optionally saved plot definition

Naming tokens

Token	Expands to	Common locations
{source}	Source worksheet name	FFT/iFFT, interpolation, statistics, fit result sheets
{column}	Processed Y/signal/value column	Formula/math/smoothing columns and result sheets where supported

Metadata role hints

Role	Typical use
X	Independent variable, time, horizontal coordinate

Y	Dependent signal/value
Z	Surface/map value
Error	Generic uncertainty fallback
X error / Y error	Errorbar and fit uncertainty defaults
Group	Statistics factor/group default
Label	Categorical/composition labels
Weight	Curve-fit weight default

Safe practices

- Save the project before opening or closing another workspace and before exiting JASC.
- Prefer unique-output policies until a batch preflight has been reviewed.
- Assign explicit Numeric/Text types when automatic inference could be ambiguous.
- Use exact-name batch binding for standardized headers and position binding only when layouts are known to match.
- Turn off Auto preview before editing expensive 2D-map settings on large datasets.
- Treat exported CSV/TXT and figures as deliverables; treat .jasc as the editable project source.